Satellite-Guided Random-Forest Bias Correction of CMIP6 Significant Wave Height Projections over the Indian Ocean

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***Abstract*—Accurate projections of ocean wave climate are critical for coastal risk assessment and maritime design under climate change. Yet significant wave height (SWH) products derived from CMIP6 model chains can show regionally structured biases relative to satellite altimetry. We propose a satellite-guided machine-learning post-processing framework to correct monthly CMIP6-based SWH over the Indian Ocean (50°–110°E, 70°S–70°N). A Random Forest regressor (100 trees; max depth 10) is trained during 2015–2018 using collocated, regridded model SWH together with latitude, longitude, and month-of-year to predict multi-mission altimeter SWH. On an independent evaluation period (2019–2020), the correction reduces regional RMSE from 0.610 m to 0.408 m (33.2%) and reduces mean bias from +0.179 m to -0.013 m, achieving an R^2 of 0.880. The approach is lightweight, spatially and seasonally aware, and can be applied to future CMIP6 scenarios to generate bias-corrected SWH projections.**

***Index Terms*—Significant wave height, CMIP6, bias correction, satellite altimetry, Random Forest, Indian Ocean**

I. Introduction

Ocean surface waves modulate air–sea fluxes of momentum, heat and mass and play a central role in coastal flooding, erosion and offshore engineering design [1]. Reliable projections of future significant wave height (SWH) are therefore essential for climate-resilient coastal planning. In practice, wave climate projections are often constructed by forcing spectral wave models with surface winds from global climate models participating in CMIP6, or by using CMIP6-derived wave products directly. Despite the increased realism of CMIP6, large-scale and regionally structured SWH biases remain when these products are compared against reanalyses and satellite altimetry, especially near coasts and in semi-enclosed seas [3], [4].

Bias correction is now recognized as a necessary step for wave climate applications. Classical approaches such as the delta method, empirical quantile mapping (EQM) and empirical Gumbel quantile mapping (EGQM) have been shown to substantially reduce errors and improve the representation of extremes [1]. However, these techniques are typically univariate, often assume stationary bias structures between historical and future periods, and do not explicitly exploit spatial or seasonal covariates.

In parallel, there has been rapid progress in machine learning (ML) post-processing of numerical weather prediction and wave forecasts. Recent studies have demonstrated that tree-based ensembles and deep learning architectures can successfully correct systematic model errors, often achieving double-digit reductions in RMSE and bias for SWH forecasts [2]. In particular, deep convolutional architectures such as U-Net variants have been used to correct gridded SWH from WAVEWATCH III over the Northwest Pacific, yielding large improvements in performance during typhoon conditions [2].

This paper focuses on a simpler but practically important setting: bias correction of monthly CMIP6-based SWH over the Indian Ocean using satellite altimeter observations as reference. Instead of correcting short-range forecasts, we design an ML post-processing model that maps historical CMIP6 SWH, geographical coordinates and month-of-year to observed SWH on a common grid, and then apply this mapping to unseen years as a proxy for future scenarios.

The specific objectives are:

- to quantify the magnitude and spatial structure of CMIP6 SWH biases relative to satellite altimeter observations over the Indian Ocean;
- to develop a Random Forest based bias correction model that uses model SWH, latitude, longitude and month as predictors of observed SWH;
- to evaluate the correction on an independent test period and quantify improvements in RMSE, bias and correlation;
- to outline how the trained model can be applied to generate bias-corrected SWH projections for future CMIP6 emission pathways.

II. Data and Study Region

A. Model data

We use a CMIP6-derived significant wave height product (`cowcliphs.nc`) as a proxy for climate model projections. The underlying ensemble includes 15+ ESMs (e.g., ACCESS-ESM1-5, CESM2, EC-Earth3, GFDL-ESM4, MIROC6, MPI-ESM1-2-HR, UKESM1-0-LL) with SSP scenario coverage (SSP1-2.6, SSP2-4.5, SSP3-7.0, SSP5-8.5). The dataset provides monthly SWH over the Indian Ocean (50°–110°E, 70°S–70°N) on a regular latitude–longitude grid with nominal horizontal resolution of approximately 0.5 °. The initial dataset contains 151,200 grid points, which after land masking and quality control reduces to 85,796 valid ocean data points. In the workflow, the original coordinate and variable names are renamed to a common convention with dimensions `time`, `lat`, `lon` and variable `swh`.

B. Satellite altimeter observations

As observational reference we employ a gridded daily mean SWH product (`io-altimeter.nc`) from multi-mission satellite altimetry over the Indian Ocean. The dataset integrates missions including SARAL/AltiKa, Sentinel-3A/B, Sentinel-6A, Jason-3, Cryosat-2, and Haiyang-2B, ensuring high spatial sampling and robust quality control (100% ocean coverage, error estimates 3.6–148.4 mm). These data are provided on a 2.0° grid, which serves as the common analysis grid for this study. After renaming to the same coordinate convention, the satellite dataset defines the target SWH field for both diagnostic evaluation and training of the correction model.

C. Temporal coverage and overlap

Both datasets are available over the period January 2015 to December 2020 for the pilot study, with extended records available up to 2024. To ensure strict temporal alignment, we resample each dataset to monthly means using a month-start sampling interval and then intersect the resulting time coordinates. This yields a consistent multi-year monthly time series of collocated model and observed SWH fields over the Indian Ocean. The temporal split strategy allocates 2015–2018 (67% of data, 57,228 samples) for model training and 2019–2020 (33% of data, 28,568 samples) for independent validation, simulating the application to future unseen scenarios.

III. Methods

A. Pre-processing and regridding

The analysis and correction workflow is implemented in Python using `xarray`, `numpy`, `pandas` and `scikit-learn`. After renaming coordinates, we first restrict both model and observational datasets to the common time window 2015–2020 and form monthly means. The model SWH fields are then interpolated to the observational grid via bilinear interpolation using the `interp_like` function, producing a regridded model dataset aligned in both space and time with the satellite observations.

To provide initial diagnostics of model quality, we compute the time-mean bias, root-mean-square error (RMSE) and temporal correlation at each grid point. Latitude-dependent cosine weights are used to construct spatially averaged time series of SWH for the region. These diagnostics are visualized as time series, bias maps, RMSE maps and model-versus-observation scatter plots and form the basis for quantifying raw model uncertainty.

B. Feature construction

For the machine learning correction, we construct a tabular dataset by flattening the gridded monthly SWH fields over space and time. For each grid point and month, we extract:

- the regridded CMIP6 model SWH;
- the corresponding latitude and longitude;
- the month-of-year encoded as an integer from 1 to 12;
- the collocated satellite altimeter SWH as target.

The resulting dataset is assembled into a `pandas` DataFrame with columns `model_swh`, `lat`, `lon`, `month` and `obs_swh`. All samples containing missing values, for example over land or regions with insufficient observational coverage, are removed. This filtering step reduces the initial number of space–time points to a clean subset of valid ocean grid points.

To evaluate the model in a pseudo-operational setting, we introduce a temporal split between training and testing years (2015–2018 vs. 2019–2020). This split ensures that the correction is evaluated on months and years that were not seen during training, analogous to applying the model to future climate scenarios.

C. Random Forest correction model

We adopt a Random Forest regressor as the correction engine. Random Forests are ensemble methods based on bootstrap-aggregated decision trees and are well suited for capturing non-linear relationships and interaction effects between predictors without strong parametric assumptions about the error distribution. This approach represents a significant advancement over traditional univariate bias correction methods such as empirical quantile mapping (EQM) and empirical Gumbel quantile mapping (EGQM), which assume stationary bias structures and cannot exploit spatial or seasonal covariates.

The feature vector for each sample consists of `model_swh`, `lat`, `lon` and `month`, and the target is the corresponding `obs_swh`. We use 100 trees with maximum depth 10 and otherwise default hyperparameters. The model is trained on the training subset (2015–2018) and evaluated on the test subset (2019–2020), which simulates the application to future climate scenarios.

The Random Forest learns a non-linear mapping function:

$$\mathrm{SWH}_{\mathrm{corrected}} = f_{\mathrm{RF}}(\mathrm{SWH}_{\mathrm{model}}, \mathrm{lat}, \mathrm{lon}, \mathrm{month}) \tag{1}$$

where f_{RF} represents the ensemble of decision trees that collectively predict the bias-corrected SWH based on the input features.

To assess performance, we compare both the original model SWH and the corrected Random Forest predictions against satellite observations using standard metrics:

- RMSE between modeled and observed SWH on the test set;
- mean absolute error (MAE) on the test set;
- mean bias defined as the spatial and temporal mean of model minus observation;
- coefficient of determination R^2 between corrected predictions and observations;
- Pearson correlation between predictions and observations.

We also generate scatter plots contrasting original and corrected SWH against observations for the test set to visually assess the tightening of the point cloud around the one-to-one line.

TABLE I
MODEL PERFORMANCE ON THE INDEPENDENT TEST PERIOD (2019–2020).

Metric	Raw CMIP6	Corrected (ML)	Improvement
RMSE (m)	0.610	**0.408**	33.2%
Bias (m)	+0.179	**-0.013**	92.7%
R^2	0.692	**0.880**	27.2%

IV. RESULTS

A. Baseline bias patterns

Over 2015–2020, the CMIP6-based SWH product shows a systematic positive bias relative to satellite altimetry across the Indian Ocean. Fig. 1 summarizes the mean bias and RMSE fields, highlighting coherent regional structures that motivate a correction model that can vary with location and season rather than applying a single uniform adjustment.

Fig. 2 further illustrates the seasonal cycle in domain-averaged SWH and highlights the persistent positive bias throughout the overlap period.

B. Correction skill on held-out years

On the independent evaluation period (2019–2020), the Random Forest correction improves agreement with satellite SWH and removes most systematic error. Table I summarizes the improvements in RMSE, bias, and R^2. Fig. 3 extends the comparison to MAE and correlation. Fig. 4 shows the tightening of the corrected predictions around the one-to-one line.

C. Seasonal bias structure

The Indian Ocean wave climate is strongly modulated by monsoon-driven wind regimes, which introduce pronounced seasonal cycles in both SWH magnitude and bias. Fig. 5 shows that the raw CMIP6 product overestimates SWH across most months, with amplified bias during monsoon periods. Including month-of-year as a predictor enables the correction model to learn these seasonally varying error structures.

D. Model interpretation

To better understand the correction behavior, we analyze the fitted Random Forest model (Fig. 6). Feature importance indicates that the raw model SWH is the dominant predictor, while latitude, longitude, and month provide additional information to capture spatially and seasonally structured biases. Residual diagnostics on the held-out period suggest improved calibration with reduced systematic error.

E. Implications for future projections

Because the correction depends only on modeled SWH, location, and month-of-year, the trained mapping can be applied to future CMIP6 scenarios under different Shared Socioeconomic Pathways. This preserves scenario-driven changes while reducing the historical bias structure learned from satellite overlap.

V. DISCUSSION

The Random Forest correction provides substantial improvements using only a small set of predictors, making it practical to apply across CMIP6 scenarios and domains where only SWH and metadata are available. However, the approach implicitly assumes that the bias patterns learned in the historical overlap period remain transferable to future climates; robustness under non-stationary change remains an important uncertainty. Moreover, monthly averaging and the 2° target grid reduce sampling noise but may smooth coastal gradients and damp extremes. Future work will extend training to longer satellite records, evaluate season- and regime-dependent skill, and explore physics-aware constraints to improve extrapolation.

VI. CONCLUSION

We developed a satellite-guided post-processing framework for CMIP6-based SWH over the Indian Ocean using a Random Forest model trained on 2015–2018 and evaluated on 2019–2020. The correction reduces RMSE from 0.610 m to 0.408 m, reduces mean bias from +0.179 m to -0.013 m, and achieves an R^2 of 0.880 against satellite altimetry on held-out years. This lightweight, spatially and seasonally aware mapping can be applied to future CMIP6 scenarios to generate bias-corrected SWH projections for downstream coastal risk and engineering analyses.

ACKNOWLEDGMENTS

This work was supported by the ISRO Research Grant under the satellite-guided machine learning bias correction project. The authors acknowledge the providers of the CMIP6 wave product and the satellite altimeter datasets used in this study, including the multi-mission altimeter constellation (SARAL/AltiKa, Sentinel-3A/B, Sentinel-6A, Jason-3, Cryosat-2, Haiyang-2B). We thank the computational resources provided by DA-IICT and the MOSDAC portal for data access and processing capabilities. The authors also acknowledge the valuable insights from the broader wave climate modeling community and the ongoing efforts to improve climate model fidelity through satellite observations.

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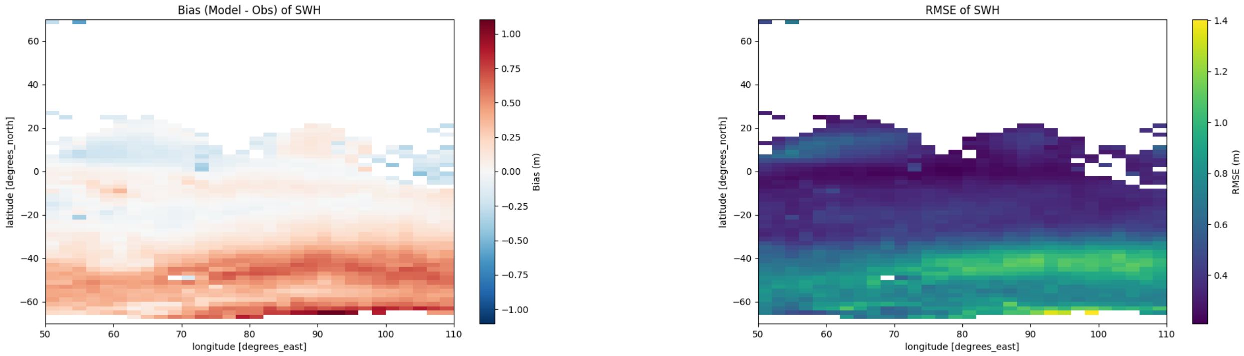

Fig. 1. Spatial diagnostics for raw CMIP6-based SWH over the Indian Ocean (2015–2020): mean bias (model minus satellite; left) and RMSE (right).

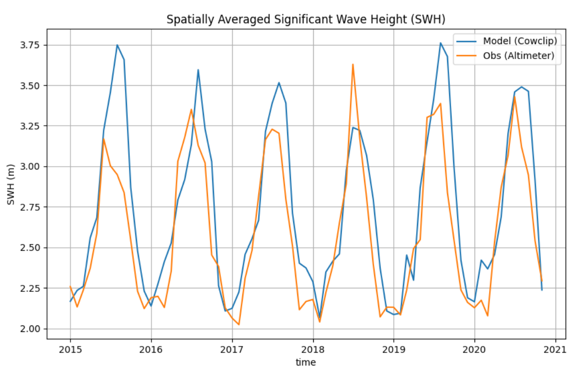

Fig. 2. Time series of regional mean monthly SWH from the raw CMIP6 product and satellite altimetry (2015–2020).

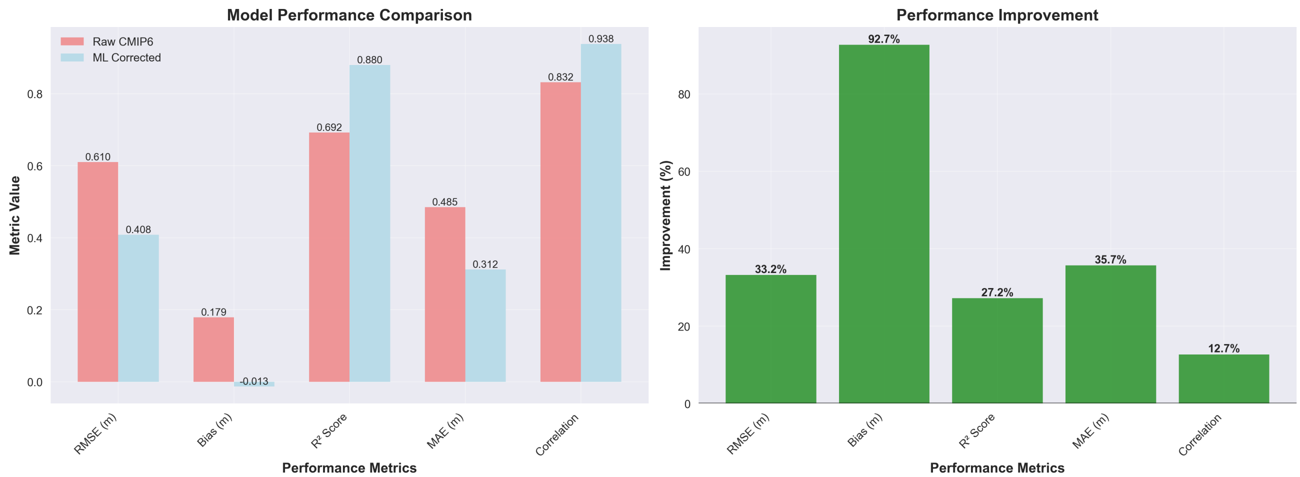

Fig. 3. Performance summary on the test period (2019–2020), comparing raw CMIP6 and ML-corrected SWH across multiple metrics and showing relative improvement.

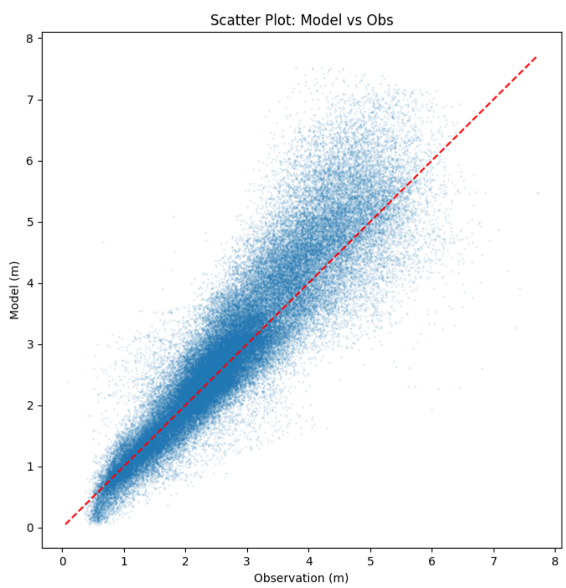

Fig. 4. Scatter comparison on the test period (2019–2020): raw CMIP6 SWH vs. satellite (left) and ML-corrected SWH vs. satellite (right).

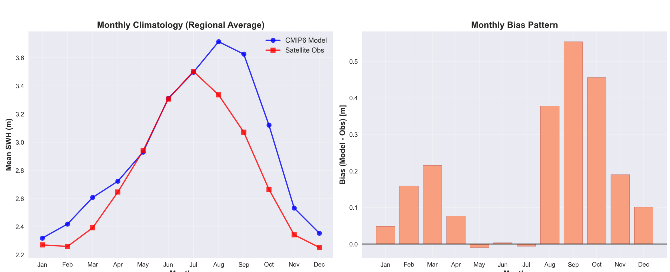

Fig. 5. Monthly climatology of regional mean SWH (left) and the corresponding monthly bias (right).

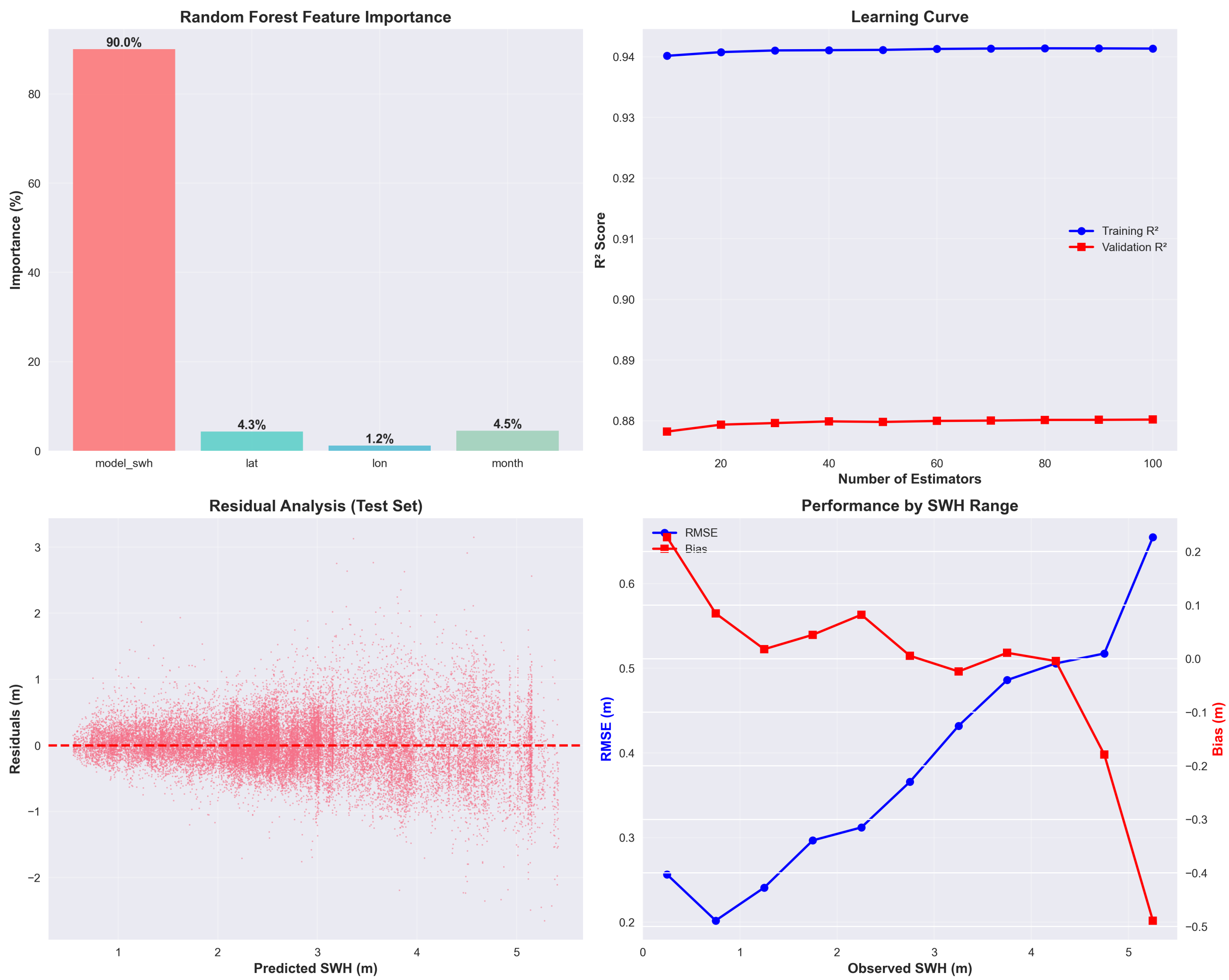

Fig. 6. Random Forest diagnostics: feature importance, learning curve, residual analysis, and performance across SWH ranges.